

Sums of two rough numbers with prescribed Liouville signs

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Abstract

Let λ be the Liouville function. We show that for every sufficiently large even integer N and each pair of signs (σ, τ) there are at least $cN/(\log N)^{1/4}$ integers $n \in [N/3, 2N/3]$ such that n and $N - n$ have no prime factor below $\exp((\log N)^{1/10})$, $\lambda(n) = \sigma$ and $\lambda(N - n) = \tau$; moreover n and $N - n$ may be taken squarefree with at most $\lceil 2 \log \log N \rceil$ prime factors. The proof combines the zero-free half-plane $\operatorname{Re} s > 7/8$ for Dirichlet L -functions, proved by OpenAI, with the weighted dilation graphs of a second OpenAI paper. The graph estimates are needed for displacements that are multiples of N and for prime groups with the prime factors of N removed, and we show that the proofs given there apply to both.

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1 Introduction

All logarithms are natural. For a positive integer n put $\lambda(n) = (-1)^{\Omega(n)}$, where Ω counts prime factors with multiplicity. Write $P^-(n)$ for its least prime factor, with $P^-(1) = \infty$, and $P^+(n)$ for its largest prime factor, with $P^+(1) = 1$. A number is z -rough if $P^-(n) > z$.

Theorem 1.1. *There are absolute constants $c > 0$ and N_0 such that, for every even integer $N \geq N_0$ and every $(\sigma, \tau) \in \{-1, +1\}^2$,*

$$\begin{aligned} \#\{n : N/3 \leq n \leq 2N/3, \quad P^-(n) > z, \quad P^-(N - n) > z, \\ \lambda(n) = \sigma, \quad \lambda(N - n) = \tau\} \geq cN/(\log N)^{1/4}, \quad z = \exp((\log N)^{1/10}). \end{aligned} \quad (1)$$

Theorem 1.2. *The representations in Theorem 1.1 may also be required to have n and $N - n$ squarefree and*

$$\Omega(n), \Omega(N - n) \leq K(N) := \lceil 2 \log \log N \rceil. \quad (2)$$

The lower bound is $cN/(\log N)^{1/4}$ with a possibly smaller absolute constant $c > 0$.

For even N the number of $n \in [N/3, 2N/3]$ satisfying just the two roughness conditions has order

$$\mathfrak{S}(N) \frac{N}{(\log z)^2} = \mathfrak{S}(N) \frac{N}{(\log N)^{1/5}}, \quad \mathfrak{S}(N) = 2 \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2}. \quad (3)$$

For completeness, the finite-residue argument in Section 6, applied just to the roughness product, gives its Euler product with an error smaller than any fixed logarithmic power. Mertens' theorem [9, Section 2.2] identifies that product with $\mathfrak{S}(N)(\log z)^{-2}$ up to absolute constant factors. The

factors of N above z change this comparison by $1 + O(L/(z \log z))$, where $L = \log N$. The positive weight used below has mass at least $cN/(\log N)^{59/250}$; we state the theorems with the exponent $1/4$.

In the author's earlier paper [4], all four sign patterns of $(\lambda(n), \lambda(N - n))$ occur $N/4 + O(N/(\log N)^c)$ times for every sufficiently large N , without a roughness restriction. Chen-type representations bound the number of prime factors in the summands without prescribing both parities. For example, Li and Liu [6] prove that every sufficiently large even N is $p + rq$, where p, q are primes, r is 1 or a prime, and $r \leq q^{0.9}$; the second summand can have either one or two prime factors. Without roughness conditions, Mangerel [7] proves under GRH the $(-1, -1)$ pattern for every sufficiently large even total and quantitative lower bounds for all four patterns at sufficiently large prime totals. CaptainSude [2] proves the $(-1, -1)$ pattern for every even $N > 2$. Meng [8, Theorem A] proves that all four patterns occur for every $N \geq 2$ outside $\{2, 3, 4, 5, 6, 9, 10\}$, and Pouly [13, Theorem 1.1] proves that, for every fixed $\varepsilon > 0$, each pattern occurs at least $c_\varepsilon N^{1-\varepsilon}$ times for every sufficiently large N . Krishnamoorthy [5] gives cancellation for the binary Liouville convolution outside a small set of totals and arbitrary logarithmic savings for its fixed higher convolutions. The sign-pattern results of [2, 5, 7, 8, 13] impose no roughness condition on the summands. A bound for sifted versions of the reflected two-point sum, $\sum_{n \leq N, P^-(n(N-n)) > z_1} \lambda(n(N-n))$ in arithmetic progressions with z_1 a fixed power of N , is stated as a hypothesis in [3, Hypothesis 3.3], and a sifted Liouville cancellation at a fixed total is formulated in [1] as the remaining analytic problem of a programme on the binary Goldbach problem. The bound $K(N)$ here grows with N .

We use the following inputs. The zero-free assertion is Input 3.1; it is [11, Theorem 1.1], the theorem of family 003 of the repository <https://github.com/openai/math>, and it is used with exactly that scope. Write [S] for [12] and [E] for [10]; both are taken from the same repository at commit

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

From [S] we use Definitions 3.1–3.3; Lemmas 3.4 and 3.6 together with their proofs; Theorem 3.5 and the residue-model moment estimate established in its proof (Sections 3.3–3.8, including Lemmas 3.7–3.10); Corollary 3.11 and its proof; Theorem 4.1 and its proof, including Lemma 4.3 and the Siegel–Walfisz estimate [S, Theorem 2.2] used for (4.9); and Lemma 4.2 with (4.6). Section 2 states the graph inputs and their order of choices. Proposition 2.5 for dilated displacements and Proposition 2.7 for punctured prime groups are obtained from the *proofs* in [S], rather than its statements. Their proofs verify, respectively, every use of the displacement in Section 3 and every use of interval prime groups in Section 4. Section 5 carries out, for the reflected operator, the physical lift and shared-label normalization of [S, Section 5.2] and the comparison reduction of [S, Section 5.3, pp. 38–39], using the sign cancellation modeled on [E, Section 4]. We do not use the arithmetic endpoint theorem [S, Theorem 5.1].

The method retains the shared prime labels in a weighted graph and then removes them using complete multiplicativity: $\lambda(D_R w) \lambda(D_R w') = \lambda(w) \lambda(w')$. Reflection is the identity $B_N = A_{-N} J_0$. The two independent free products in each comparison term produce rational frequency grids with denominators Nr . Character orthogonality retains the factor $1/Q$ at $Q = Nr$; the zero-free input bounds the short nonprincipal prime sum, and a small-prime modification bounds the principal residual mean. The residual length exceeds Q in the comparisons. At the raw scale $H' = 1$ this inequality fails, so the grid estimate is used only for comparison terms.

The exponent $1/10$ is not the limit of these estimates. The rough cutoff must lie below the group primes, and the group exponents satisfy $d < 0.47$ in [S]. This bound keeps the residue moduli of moment length $L^{0.52}$ equal to $\exp(o(L))$, and it prevents the argument from reaching $z = N^\delta$ for

fixed $\delta > 0$. The threshold N_0 is not effective with the Siegel–Walfisz input used in [S, Section 4]. We make no assertion for a fixed number of prime factors or for prime summands.

Section 2 gives the graph inputs, dilation and reflection. Section 3 establishes the character estimates. Section 4 proves the grid energy estimate, including a partial-summation verification of the smooth weights. Section 5 proves reflected cancellation. Section 6 constructs the positive weight and its mass. Section 7 deduces the two theorems.

2 Graph inputs, dilation and reflection

We use $L = \log x$ and $T = \log L$ in this section.

Definition 2.1 (Prime groups and patterns, [12, Definitions 3.1–3.3]). Fix the data

$$0 < a < b < c < d < 0.47, \quad c_0, C_0, v_-, v_+ > 0, \quad 0 < q_0 < 1, \quad \ell \in \mathbb{N}_{>0}. \quad (4)$$

Let $\mathcal{P}_g$ be finite pairwise disjoint prime sets indexed by disjoint small and big families $\mathcal{G}_{\text{small}}, \mathcal{G}_{\text{big}}$. Each family has between $c_0 T$ and $C_0 T$ groups. Put

$$\begin{aligned} \exp(L^a) \leq p \leq \exp(L^b) \quad (g \in \mathcal{G}_{\text{small}}), \quad \exp(L^c) \leq p \leq \exp(L^d) \quad (g \in \mathcal{G}_{\text{big}}), \\ v_- \leq V_g := \sum_{p \in \mathcal{P}_g} \frac{1}{p} \leq v_+, \quad \mu_g(p) = \frac{1}{pV_g}, \quad \omega_g(n) = \sum_{p \in \mathcal{P}_g} \mathbf{1}_{p|n}, \\ \mathcal{P} = \bigcup_g \mathcal{P}_g, \quad s = |\mathcal{G}_{\text{small}}| + |\mathcal{G}_{\text{big}}|, \quad \omega_{\mathcal{P}}(n) = \sum_g \omega_g(n). \end{aligned} \quad (5)$$

Divisibility at every $n \in \mathbb{Z}$, including 0, has its usual meaning. For a fixed positive integer J set $M = J + \ell$. A pattern ν chooses $C_g \subset \{1, \dots, J\}$, empty in small groups and of size at most a fixed m in big groups. Put $t_g = \ell + |C_g|$. The slots in $\{1, \dots, J\} \setminus C_g$ are shared between the ordered source and target lists. The other t_g endpoint slots are unshared. Independent auxiliary labels with laws μ_g fill the C_g slots in the edge product D , which uses J prime factors in each group. Write $D = D_{\text{small}} D_{\text{big}}$. The coefficients depend only on ordered big source, big target and auxiliary big labels, and satisfy

$$\sum_{\nu} \sup |K_{\nu}| \leq L^{C_1}, \quad (6)$$

where m, C_1 are independent of J . Every operator is symmetrized on both sides by averaging the $M!$ slot permutations independently in each group.

Definition 2.2 (Ideal and physical operators). The ideal Hilbert space is the product probability space of M big-group labels with law μ_g in every big group; numerical repetitions are allowed. A row retains shared labels, independently draws unshared target and auxiliary free labels, and multiplies by

$$K_{\nu} D_{\text{big}}^{i\zeta} e(\Theta D_{\text{big}}), \quad \Theta, \zeta \in \mathbb{R}, \quad e(t) = \exp(2\pi i t). \quad (7)$$

Sum and symmetrize to obtain $T_{\zeta}(\Theta)$. A physical state is $(n, \mathbf{p})$, with $n \in \mathbb{Z}$ and M ordered distinct labels from each group, all dividing n . Its mass is $\prod_g V_g^{-M}$, with counting measure in n . For an integer $0 < |k| \leq L^{C_2}$ the row moves to $n + kD$, sums compatible target lists with coefficient $\prod_g V_g^{-t_g}$, and draws free labels with their μ_g laws. Each auxiliary free label is forbidden from both endpoint lists, although free labels may repeat one another. Its multiplier is

$$K_{\nu} D^{i\zeta} q_0^{(\omega_{\mathcal{P}}(n) - Ms)/2} q_0^{(\omega_{\mathcal{P}}(n + kD) - Ms)/2}. \quad (8)$$

Call the symmetrized sum A . Its adjoint reverses the displacement and interchanges the endpoint labels, conjugating the multiplier. The joint list normalization is $V_g^{-M-t_g}$ in either direction. The vector u_n equals 1 on all lists at position n and 0 elsewhere; it is not normalized.

Input 2.3 (Absolute bounds and transference, [12, Lemma 3.4, Theorem 3.5, Corollary 3.11]). *The absolute row and column sums of A are bounded by $L^{C_{\text{abs}}}$, with C_{abs} independent of J (it may depend on m, C_1 and the common fixed data); this also holds on the independent-residue position space. For each fixed $E > 0$ there is $E_{\text{id}} > 0$, depending only on E and the common fixed data, such that*

$$\sup_{\Theta} \|T_{\zeta}(\Theta)\| \leq L^{-E_{\text{id}}} \quad (9)$$

implies, for every sufficiently large fixed J , with $R = \lceil L^{0.52} \rceil$, every fixed $\gamma > 0$ and every positive dyad $I = [Y, 2Y)$, $x^\gamma \leq Y \leq x^{1/\gamma}$,

$$\frac{1}{|I \cap \mathbb{Z}|} \sum_{n \in I \cap \mathbb{Z}} \langle u_n, (AA^*)^R u_n \rangle \leq L^{-2RE}. \quad (10)$$

The lower bound on J may depend on m, C_1 ; the threshold in x may depend on all fixed parameters, including J, C_2, γ . Subject to these data and (9), the estimate is uniform in groups, patterns, k and ζ . There is no further bound on ζ when the ideal hypothesis holds. If f is supported on I , is independent of its list of marks, $\sup |f_n| \leq \exp(C_f \sqrt{L})$, and $\|f\|, \|g\| \leq \sqrt{Y} L^{C_3}$, then

$$|\langle f, Ag \rangle| \ll Y L^{-E+2C_3}. \quad (11)$$

Only the first endpoint must be independent of its mark list.

Input 2.4 (Residual ideal kernels, [12, Theorem 4.1, with (4.7)–(4.8) and the last two paragraphs of its proof, p. 34]). *Suppose initially that each big group is the set of all primes in an interval in its permitted range. For each fixed $E_{\text{id}} > 0$ and $C_4 \geq 0$ there are fixed m, J_{min}, C_1 and kernel parameters, independent of J , such that every fixed $J \geq J_{\text{min}}$ admits one raw pattern $C_g = \emptyset, K_0 = 1$ and signed comparison patterns satisfying*

$$\sup_{\Theta \in \mathbb{R}, |\zeta| \leq L^{C_4}} \|T_{\zeta}(\Theta)\| \leq L^{-E_{\text{id}}}. \quad (12)$$

The family and coefficients are independent of the particular Θ, ζ . Split the big groups into two blocks and designate m of the first J slots in each big group as probes. Each comparison frees nonempty probe sets $\mathcal{A}, \mathcal{B}$ in the two blocks, with coefficient

$$-(-1)^{|\mathcal{A}|+|\mathcal{B}|} K_{\mathcal{A}}(D_{\mathcal{A}}, X_{\mathcal{A}}) K_{\mathcal{B}}(D_{\mathcal{B}}, X_{\mathcal{B}}). \quad (13)$$

Here $D_{\mathcal{A}}, D_{\mathcal{B}}$ are independent free-label products; $X_{\mathcal{A}}$ uses target labels and $X_{\mathcal{B}}$ source labels. No shared label or final ℓ label enters these coefficients. For a slot set Λ ,

$$K_{\Lambda}(D, X) = \sum_{j: \nu_j \geq \xi} \frac{\mathbf{1}_{\log D \in I_j} \mathbf{1}_{\log X \in I_j}}{\nu_j} \sum_{\substack{\chi \text{ primitive} \\ \text{cond}(\chi) \leq Q_0}} \overline{\chi(D)} \chi(X), \quad (14)$$

where $I_j = [j\mathfrak{h}, (j+1)\mathfrak{h})$ has width $\mathfrak{h} \leq 1$, ν_j is the product-law cell probability, and $\mathfrak{h}^{-1}, \xi^{-1}, Q_0$ are fixed powers of L . The conductor-1 character is included. The number and total coefficient mass of the expanded terms are fixed powers of L , independent of J . Every free product contains a big prime.

The order in these inputs is substantive. In [S, (3.37), p. 23] one fixes $G > E + C_v + 3$ and $E_{\text{id}} > G + C_s + 2$ before increasing J . In [S, pp. 33–34] one fixes the probe count m , then the comparison accuracy, then the kernel parameters in (4.20) and their coefficient exponent C_1 , and only then imposes a lower bound on J . Thus J may subsequently be increased to satisfy transference without changing any kernel precision. We will keep this order.

Proposition 2.5 (Coprime integer dilation). *Keep Definitions 2.1–2.2 and the fixed bound $0 < |k| \leq L^{C_2}$. Let $h \neq 0$ be any integer coprime to every group prime, and replace every physical displacement kD by hkD , leaving the coefficients and all mark rules unchanged. Call this operator A_h . Under (9), with the same order of fixed parameters, one has, for every fixed $\gamma > 0$ and every positive dyad $[Y, 2Y)$ with $x^\gamma \leq Y \leq x^{1/\gamma}$,*

$$\sum_{Y \leq n < 2Y} \langle u_n, (A_h A_h^*)^R u_n \rangle \ll Y L^{-2RE} \quad (15)$$

and, under the endpoint hypotheses of Input 2.3,

$$|\langle f, A_h g \rangle| \ll Y L^{-E+2C_3}. \quad (16)$$

The absolute row and column bounds of Input 2.3 hold for A_h with the same C_{abs} , also on the independent-residue position space. The constants and threshold are independent of h and its size. This proposition is deduced from the proofs of Lemmas 3.4 and 3.6, the residue-model moment estimate in the proof of Theorem 3.5, and the proof of Corollary 3.11 of [S].

Proof. We spell out the dependence on the displacement in [S, Section 3]; in this proof, Definition, Lemma, Theorem and Corollary numbers and equation numbers of the form (3. n) are those of [S]. Definition 3.3 and the adjoint computation (3.3)–(3.4), pp. 11–12, replace $n + kD$ by $n + hkD$; reversal still keeps D fixed and the joint normalization unchanged. The absolute estimate (3.5) in Lemma 3.4, p. 12, counts target marks and uses only divisibility and translation of the position; its row and column bounds therefore hold for A_h , also on the independent-residue position space. The range of k in Theorem 3.5, p. 12, is the restriction being extended; its residue proof uses that restriction only in the ways checked below.

In the root-path expansion, (3.8), p. 13, the offsets become hb_i , where $b_0 = 0$ and $b_{i+1} - b_i = \pm kD_i$. Return is $hb_{2R} = 0$, equivalent to $b_{2R} = 0$. Thus the closed numerical paths and scalar coefficients are exactly those for $h = 1$. Lemma 3.6 and (3.9)–(3.10), pp. 13–14, specify congruences $n + hb_i \equiv 0 \pmod{p}$. Their active and Bonferroni moduli use the same primes, have size $\exp(o(L))$, and each CRT count on $[Y, 2Y)$ has rounding error $O(1)$ independently of the representatives hb_i . The closure phase (3.11), p. 14, can be kept as $e(\theta b_{2R})$, since the return indicators coincide.

In the independent-residue law the measure-preserving map $n_p \mapsto h^{-1}n_p$ changes every predicate $n_p + hb_i = 0$ to $n_p + b_i = 0$. It preserves active visits, repeated residues and all damping. Hence every numerical-path expectation, including the absolute sum that controls replacement error, is exactly the $h = 1$ expectation. Consequently the signed residue-model path sum for A_h , with the return condition $b_{2R} = 0$, equals the one for $h = 1$, which the proof of [S, Theorem 3.5] bounds by $o(L^{-2RE})$ (Sections 3.3–3.8 and the completion on p. 25). The remainder of this paragraph and the next two paragraphs record that each step of that proof also applies directly to the displacement hkD . The lifespan tests (3.13)–(3.14), pp. 14–15, are preserved as well. The assertion on p. 15 that every group prime exceeds $|k|$ is used to infer $p \mid kD \Rightarrow p \mid D$. For the enlarged displacement, $p > |k|$ and $p \nmid h$ give $p \mid hkD \Rightarrow p \mid D$. Its converse exclusion for unshared entry and exit labels is therefore unchanged.

In the edge-memory operations on p. 16, the translation $\Delta = \pm kD$ becomes $\Delta = \pm hkD$, while the phase remains $e(\pm \theta kD)$ when the closure phase is kept as $e(\theta b_{2R})$. Translations remain measure preserving. The exact memory identity (3.17), p. 17, uses only these congruences and the just-verified active/shared implication. The reversed memory measures (3.23)–(3.25), p. 19, remain unchanged under any translation Δ ; their absolute bounds, (3.26)–(3.28), p. 20, have no size dependence. For rare transfers, (3.31) and Lemma 3.9, pp. 20–21, the congruence is $r + \sigma_i hk D_{\text{big}} F/U = 0 \pmod{p}$. The omitted small-label products satisfy $U \leq \exp(\ell |\mathcal{G}_{\text{small}}| L^b) < \exp(L^c) \leq p$; they are distinct modulo p , and h, k, F, D_{big} are invertible there whenever $p \nmid D$. At most one omission choice is allowed, exactly as in the source.

The clean-edge Fourier reduction (3.36) and Lemma 3.10, pp. 21–22, only change the real frequency Θ in $e(\sigma_i \Theta D_{\text{big}})$. The ideal hypothesis is uniform in every real Θ . The equality-rank expansion (3.39)–(3.47), pp. 23–25, changes no birth labels, point masses or equality phases; its unaffected edges retain this same clean estimate. The completion on p. 25 thus gives the same residue-model moment. Restoring the uniform residue error with a slightly stronger fixed precision proves (15). This is a full-root CRT average, with no assertion of residue independence along a short progression modulo h .

It remains to adapt the physical interval bound H in [S, (3.49), p. 26]. Choose

$$H_0 \geq \max |kD|, \quad \log H_0 = O_J(TL^d). \quad (17)$$

For each residue class modulo $|h|$, partition its integer quotient coordinate into blocks of $\lceil H_0 \rceil$ consecutive positions. Write block_j for the part of one such block in $[Y, 2Y)$, let f_j be the restriction of f to block_j , and let g_j be the restriction of g to the union of the full quotient block containing block_j and the two adjacent full quotient blocks of that same residue class, without intersecting these blocks with $[Y, 2Y)$. Every edge stays in the class and moves at most H_0 quotient positions. Therefore

$$\langle f, A_h g \rangle = \sum_j \langle f_j, A_h g_j \rangle, \quad \sum_j \|g_j\|^2 \leq 3 \|g\|^2. \quad (18)$$

The positive semidefinite matrix $(\langle u_n, (A_h A_h^*)^R u_m \rangle)_{n,m \in \text{block}_j}$ has largest eigenvalue at most its trace d_j . Mark independence gives

$$\langle f_j, (A_h A_h^*)^R f_j \rangle \leq \left(\sum_{n \in \text{block}_j} |f_n|^2 \right) d_j \leq \lceil H_0 \rceil \sup |f_n|^2 d_j. \quad (19)$$

Spectral Hölder and Hölder in j , with exponents $2, 2R/(R-1), 2R$, give

$$|\langle f, A_h g \rangle| \leq \sqrt{3} \|g\| \|f\|^{1-1/R} (\lceil H_0 \rceil \sup |f_n|^2)^{1/(2R)} \left(\sum_j d_j \right)^{1/(2R)}. \quad (20)$$

The middle factor is bounded, since $d < 0.47 < 0.52$ and $0.5 < 0.52$. The blocks partition the support dyad, so their traces sum to (15). The norm hypotheses prove (16). Empty blocks, including the case $|h| > 2Y$, cause no change. $\square$

Proposition 2.6 (Reflection). *Let $(J_0 g)(n, \mathbf{p}) = g(-n, \mathbf{p})$. For $h \neq 0$ coprime to the group primes, let B_h have the source mark transitions with physical target $hkD - n$. Then*

$$B_h = A_{-h} J_0, \quad B_h B_h^* = A_{-h} A_{-h}^*. \quad (21)$$

Consequently (16) and the absolute row and column bounds of Proposition 2.5 hold with B_h in place of A_h .

Proof. Reflection preserves divisibility, state measure and degree damping, so J_0 is a unitary permutation of the states and leaves absolute row and column sums unchanged. The translation target $n - hkD$ becomes $hkD - n$ after applying J_0 , with exactly the same admissible lists. The squared-operator identity follows from $J_0 J_0^* = 1$. The second endpoint in the translation interface may be supported at negative integers, since the physical space is all of $\mathbb{Z}$. We only use Liouville values at the positive endpoints before applying J_0 . $\square$

Proposition 2.7 (Punctured prime intervals). *Set $x = N$. Remove the primes dividing N from each prime interval used for a big group. If the resulting groups satisfy (5), the conclusion and parameter order of Input 2.4 hold for them. The transference inputs already allow arbitrary finite groups.*

Proof. For any fixed $\alpha > 0$, deletion has reciprocal mass at most

$$\sum_{\substack{p|N \\ p \geq \exp(L^\alpha)}} \frac{1}{p} \leq \frac{L}{L^\alpha \exp(L^\alpha)}. \quad (22)$$

Indeed at most L/L^α distinct prime factors can lie above this threshold. This is smaller than every fixed negative power of L , also after any further interval intersection. For big groups take $\alpha = c$; $\alpha = a$ controls both kinds of groups.

Here is the full check of the interval uses in [S, Section 4]; in this proof, Theorem, Lemma and equation numbers of the form (4.n) are those of [S]. Its opening assumption on p. 27 says that intersecting a big group with another interval is still an interval of primes. After deletion it is that interval with only primes dividing N missing. For a selected slot set Λ , write P_Λ for its label product, k_g for its number of slots in group g , and a_p for the multiplicity of p in a supported product n . Let $\rho_\Lambda(n) = \mathbb{P}(P_\Lambda = n)$, zero outside the support. In (4.3), p. 28, unique factorization gives, on this support, the mass

$$\rho_\Lambda(n) = \frac{1}{n} \prod_g \frac{k_g!}{V_g^{k_g} \prod_{p \in \mathcal{P}_g} a_p!} \leq \frac{L^\kappa}{n}, \quad \kappa = C_0 \{\log(m!) + m \log \max(1, v_-^{-1})\}, \quad (23)$$

for at most m selected slots per big group. This uses no interval distribution and is unchanged, including repeated labels. The fiber Cauchy bound (4.4), p. 28, is

$$\sum_n |\mathbb{E}[f \mathbf{1}_{P_\Lambda=n}]|^2 \leq \frac{L^\kappa}{c_1 U} \|f\|_2^2 \quad (c_1 U \leq P_\Lambda \leq c_2 U); \quad (24)$$

it follows from the same mass bound for arbitrary tuple tests.

The log-cell definitions (4.7) and bounds (4.8), p. 29, become

$$N_\Lambda \ll_m 1 + TL^d/\mathfrak{h}, \quad \sup |K_\Lambda| \leq Q_0^2 \xi^{-1}, \quad \sup_X \mathbb{E}_D |K_\Lambda(D, X)| \leq Q_0^2. \quad (25)$$

Support size, retained cell mass and the cancellation of ν_j give these inequalities for the new product law.

The only distribution estimate is (4.9), pp. 29–30:

$$\left| \mathbb{E} \left[\mathbf{1}_{\log D \in I_j} \overline{\chi(D)} \chi'(D) \right] \right| \ll_{D^*} L^{-D^*} \quad (\chi \neq \chi'). \quad (26)$$

Fix all but one selected label. The remaining prime lies in the intersection of its original interval and a cell-imposed interval. The character $\bar{\chi}\chi'$ is nonprincipal on a modulus at most Q_0^2 , by uniqueness

of primitive induction. The reciprocal prime sum used in [S, (4.9)] is valid on the full intersection by [S, Theorem 2.2]; removing $p \mid N$ changes it by at most (22). Division by $V_g \geq v_-$ and averaging the other labels preserve arbitrary precision. Every remaining prime exceeds Q_0^2 , so all conditional diagonal Gram entries in (4.10), p. 30, are exactly 1. Thus (4.11), p. 30, retains the bound

$$1 + O_{D^*}(Q_0^2 \xi^{-1} L^{-D^*}) \leq 2. \quad (27)$$

The character-reproduction formula (4.18), p. 32, remains

$$\mathbb{E}_D K_\Lambda(D, X) \psi(D) = \psi(X) + O_{D^*}(Q_0^2 \xi^{-1} L^{-D^*}) \quad (28)$$

on retained cells: its diagonal term is exact and the others use (26). Applying it twice with the rational character expansion (4.15) preserves (4.19), p. 32, whose error is

$$O_{D^*}(Q_0^{5/2} \xi^{-1} L^{-D^*}). \quad (29)$$

For the remaining displays, the integer bilinear estimate (4.5)–(4.6), pp. 28–29, permits coefficients vanishing on any subset of containing intervals. The cell support comparison and derivative error (4.13)–(4.17), pp. 30–31, use only product sizes, $\mathfrak{h}$, and the new cell probabilities; discarded cells still have total mass at most $(N_{\mathcal{A}} + N_{\mathcal{B}})\xi$. The parameter inequalities (4.20), p. 32, absorb (22) at any fixed precision before J . The symmetric-probe fractions and cancellations (4.21)–(4.26), pp. 32–34, use a product probability law and within-group permutation invariance, both of which persist. These are all the interval dependencies: none asserts distribution for an arbitrary subset of a prime interval. The proof of [S, Theorem 4.1] therefore gives the proposition, uniformly in N . $\square$

3 Zero-free input and character estimates

Input 3.1 (Family 003, [11, Theorem 1.1]). *For every positive integer q , every Dirichlet character $\chi \pmod{q}$ and every $s \in \mathbb{C}$ with $\operatorname{Re} s > 7/8$, one has $L(s, \chi) \neq 0$, apart from the principal character's pole at $s = 1$.*

The Lean declaration is

```
OAI.DirichletCharacter.LFunction_ne_zero_of_seven_eighths_lt_re.
```

The companion declaration `OAI.riemannZeta_ne_zero_of_seven_eighths_lt_re` gives the zeta assertion away from its pole.

Lemma 3.2 (Liouville character power bound). *For fixed $C_1 > 0$ and every fixed $\alpha > 7/8$,*

$$\sum_{n \leq X} \lambda(n) \chi(n) n^{it} \ll_{C_1, \alpha} X^\alpha \quad (q, |t| \leq X^{C_1}), \quad (30)$$

uniformly in all characters modulo q , including principal and imprimitive characters.

Proof. We give the analytic details, using the classical Borel–Carathéodory and three-lines inequalities and the Dirichlet-series and contour tools of [9, Chapters 5, 10–13]. For primitive nonprincipal χ , Input 3.1 permits a holomorphic branch $F(s) = \log L(s, \chi)$ on $\operatorname{Re} s > 7/8$, agreeing with the Euler logarithm on $\operatorname{Re} s > 1$. Partial summation of the character sum, bounded by q , gives $|L(s, \chi)| \leq q|s|/\operatorname{Re} s$ for $\operatorname{Re} s > 0$. Borel–Carathéodory on discs centered at $2 + it$, staying a fixed

distance to the right of $7/8$, bounds $|F(7/8 + \delta/2 + it)| \ll_\delta \log(q(|t| + 3))$. On $\operatorname{Re} s = 1 + \delta/2$ the Euler logarithm is $O_\delta(1)$. Apply three-lines to $F(s) \exp((s - it_0)^2/\mathcal{R}^2)$, where $\mathcal{R} = q + |t_0| + 3$. The Gaussian gives finite boundary suprema. At $s = \sigma + it_0$, $\sigma \geq 7/8 + \delta$, this yields

$$|F(s)| \ll_\delta [\log(q(|t_0| + 3))]^{1-\eta_\delta}, \quad \eta_\delta > 0, \quad (31)$$

and hence, for every $\varepsilon > 0$,

$$L(s, \chi)^{-1} \ll_{\delta, \varepsilon} (q(|t| + 3))^\varepsilon \quad (\operatorname{Re} s \geq 7/8 + \delta). \quad (32)$$

For the primitive principal character use the nonvanishing analytic function $Z(s) = \zeta(s)(s-1)/(s+1)$, whose canceled value at 1 is nonzero. Its logarithm has the same bounds; the factor $(s-1)/(s+1)$ in $1/\zeta(s)$ is bounded by 1 on $\operatorname{Re} s \geq 0$. For imprimitive characters, the inverse local-factor logarithm is $O_\delta(\sum_{p|q} p^{-7/8-\delta}) = o(\log q)$. To see uniformity of the last assertion, fix a large prime threshold: the finitely many smaller primes cost $O(1)$, and the contribution of larger primes, divided by $\log q$, is arbitrarily small. It is absorbed by another arbitrarily small conductor power.

The twisted series is

$$\sum_{n \geq 1} \frac{\lambda(n) \chi(n) n^{it}}{n^s} = \frac{L(2s - 2it, \chi^2)}{L(s - it, \chi)}. \quad (33)$$

The numerator is absolutely bounded on a fixed line right of $7/8$. Truncated Perron at a half-integer cutoff and height $\mathcal{T}$ has error $O(X \log^2 X/\mathcal{T} + 1)$. Move to $\operatorname{Re} s = 7/8 + \delta < \alpha$; the vertical term is

$$O\left(X^{7/8+\delta} (q(\mathcal{T} + |t| + 3))^\varepsilon \log \mathcal{T}\right), \quad (34)$$

and the horizontal integrals are smaller after a sufficiently large fixed-power height is chosen. There is no principal pole in the ratio: the pole of $L(s - it, \chi)$ becomes a zero of its inverse. Choose the reciprocal-growth exponent ε after C_1 , the height exponent and the margin $\alpha - 7/8 - \delta$. This proves (30). The same reasoning applies when $q, |t| \leq N^C$ and $N^\eta \leq X \leq N^{1/\eta}$ by enlarging C_1 . $\square$

Lemma 3.3 (Small-prime modifications). *Fix $C > 0$, $0 < d < 1$, and $L = \log N$. Let $Z_0 \leq \exp(L^d)$ and let f be real and completely multiplicative, possibly depending on N , with $f(p) \in [-1, 0]$ for $p \leq Z_0$ and $f(p) = -1$ otherwise. For every fixed $\eta, A > 0$,*

$$\sup_{N^\eta \leq X \leq N^{1/\eta}} \sup_{\substack{q \leq N^C \\ |t| \leq N^C}} \sup_{\chi \bmod q} \left| \sum_{n \leq X} f(n) \chi(n) n^{it} \right| \ll_{C, d, \eta, A} X L^{-A}. \quad (35)$$

This includes every character and real t ; the same bound holds on any subinterval of $[X, 2X]$.

Proof. Write $f(p) = -u_p$, $0 \leq u_p \leq 1$. The exact convolution $f = \lambda * g$ has local coefficients

$$\sum_{j \geq 0} g(p^j) v^j = \frac{1 + v}{1 + u_p v}, \quad g(p^j) = (1 - u_p)(-u_p)^{j-1} \quad (j \geq 1) \quad (36)$$

at modified primes, and $g(p^j) = 0$ at all others. For $0 < v < 1$,

$$\sum_{j \geq 1} |g(p^j)| v^j = \frac{(1 - u_p)v}{1 - u_p v} \leq v. \quad (37)$$

Mertens' estimates give $\sum_n |g(n)|/n \ll \log(2Z_0)$. For large Z_0 , take $\delta = 1/\log Z_0$ and apply Rankin:

$$\sum_{n > \sqrt{X}} \frac{|g(n)|}{n} \leq X^{-\delta/2} \prod_{p \leq Z_0} (1 + p^{-1+\delta}) \ll X^{-1/(2 \log Z_0)} (\log(2Z_0))^e. \quad (38)$$

The last estimate uses $p^\delta \leq e$ and the harmonic prime form of Mertens' theorem. For $n \leq \sqrt{X}$, Lemma 3.2 with $\alpha = 15/16$ gives

$$X^{15/16} \sum_{n \leq \sqrt{X}} |g(n)| n^{-15/16} \leq X^{31/32} \sum_n |g(n)|/n.$$

Thus the complete sum is

$$\ll X^{31/32} \log(2Z_0) + X \exp\left(-\frac{\log X}{2 \log Z_0}\right) (\log(2Z_0))^e. \quad (39)$$

Both terms are $O_A(XL^{-A})$, uniformly in u_p, χ, t . Bounded Z_0 uses a fixed Rankin exponent; no modifications gives $f = \lambda$ directly. Differences of prefixes give subintervals. $\square$

Corollary 3.4 (Rough one-factor means). *Put $R_z(n) = \mathbf{1}_{P^-(n) > z}$. For fixed $C_0 > 0$, $0 < \rho_0 < 1$ and $H > 0$, uniformly for $2 \leq z \leq \exp((\log X)^{\rho_0})$, $q, |t| \leq X^{C_0}$ and all characters modulo q ,*

$$\sum_{n \leq X} \lambda(n) R_z(n) \chi(n) n^{it} \ll_{C_0, \rho_0, H} X (\log X)^{-H}. \quad (40)$$

Proof. Let μ be the Möbius function, equal to 0 on nonsquarefree integers and to $(-1)^{\omega(n)}$ on squarefree integers, where $\omega(n)$ counts distinct prime factors. The coefficient identity, including prime powers, is

$$\lambda R_z = \lambda * g_z, \quad g_z(d) = \mu^2(d) \mathbf{1}_{P^+(d) \leq z}. \quad (41)$$

At $p \leq z$ the factor $1 + p^{-s}$ cancels the Liouville Euler factor; at other primes it is 1. Writing $\alpha = 15/16$ and taking the conductor exponent $C_1 = 2C_0$ on the short convolution gives

$$\sum_{d \leq \sqrt{X}} g_z(d) (X/d)^\alpha \leq X^{(1+\alpha)/2} \sum_d g_z(d)/d \ll X^{31/32} \log(2z). \quad (42)$$

With $\varepsilon_z = 1/\log z$ the other part satisfies

$$\sum_{d > \sqrt{X}} \frac{g_z(d)}{d} \leq X^{-1/(2 \log z)} \prod_{p \leq z} (1 + p^{-1+1/\log z}) \ll X^{-1/(2 \log z)} (\log(2z))^e. \quad (43)$$

The finite product is valid even if $\varepsilon_z > 1$ for small fixed z . The total bound is

$$C \left\{ X^{31/32} \log(2z) + X \exp\left[-\frac{\log X}{2 \log z}\right] (\log(2z))^e \right\}. \quad (44)$$

Since $\log z \leq (\log X)^{\rho_0}$ with $\rho_0 < 1$ fixed, both terms are $O_H(X(\log X)^{-H})$, which proves (40). $\square$

Lemma 3.5 (Short prime character sums). *For fixed $C > 0$, $0 < a < b < 1$, and $\exp(L^a) \leq P \leq \exp(L^b)$, every nonprincipal character modulo $q \leq N^C$, every real $|t| \leq N^C$, and every interval $I \subset [P, 2P]$ satisfy*

$$\left| \sum_{p \in I} \chi(p) p^{it} \right| \ll_C P^{15/16} L^{C_1}, \quad (45)$$

where C_1 is fixed (with the fixed exponent ranges understood). The same estimate holds after omitting primes dividing N . For a principal character we use only $O(P)$.

Proof. Cauchy's formula for the analytic logarithm in Lemma 3.2, or its standard zero partial-fraction expansion, gives $-L'/L(s, \chi) = O(\log^2(q(|\Im s| + 3)))$ on $\operatorname{Re} s = 15/16$ and the horizontal segments, for nonprincipal characters. Apply truncated Perron to $-L'/L(s - it, \chi)$, initially with height N^{C+10} , increasing its fixed exponent if required. Shift to $\operatorname{Re} s = 15/16$. The pole at $1 + it$ is absent precisely for nonprincipal χ . The zero-free input gives a fixed distance from all zeros. The remaining line contributes $O(P^{15/16} L^{C_1})$. Truncation and horizontal errors are polynomially small after the height is enlarged, even for $P = \exp(L^a)$. Proper prime powers cost $O(P^{1/2} L^{C_1})$. Partial summation removes $\log p$; differences of prefixes treat I . Imprimitive Euler factors and deletion of $p \mid N$ contribute at most a fixed power of L , absorbed in (45). Thus the modulus need only be polynomial in N , not in P . A principal pole would supply a main term and is never discarded. $\square$

4 Fourier energy on rational grids

From now on $L = \log N$. Fix the separated ranges

$$\begin{aligned} z &= \exp(L^{1/10}), \quad \mathcal{P} \subset [\exp(L^{0.12}), \exp(L^{0.18})] \cup [\exp(L^{0.35}), \exp(L^{0.40})], \\ \mathcal{S} &= \{p : \exp(L^{0.22}) < p \leq \exp(L^{0.28}), p \nmid N\}. \end{aligned} \quad (46)$$

Assume $\mathcal{P}$ contains no prime factor of N and $\sum_{p \in \mathcal{P}} 1/p = O(\log L)$. Write $n = n_{\mathcal{P}} n_*$, where $n_{\mathcal{P}}$ is the full $\mathcal{P}$ -part, including prime powers. Use $\omega_{\mathcal{S}}$ for the number of distinct $\mathcal{S}$ -prime divisors and $\Omega_{\mathcal{S}}$ for their number with multiplicity.

Lemma 4.1 (Grid energy). *Let $c(g)$ be a complex function supported on $\mathcal{P}$ -smooth integers, that is, integers all of whose prime factors lie in $\mathcal{P}$, and bounded by L^C . It may depend on N and all fixed comparison parameters. Define*

$$a(n) = \psi(n/Y) n^{iv} \lambda(n_*) R_z(n_*) \mathbf{1}_{(n, N)=1} \mathbf{1}_{\omega_{\mathcal{S}}(n_*) > 0} c(n_{\mathcal{P}}), \quad (47)$$

where ψ is supported on a fixed compact subinterval of $(0, \infty)$, $\|\psi^{(j)}\|_{\infty} = O_j(L^{C(j+1)})$, and $v \in \mathbb{R}$, $|v| \leq L^C$. Fix $C_0, C_2 > 0$. Uniformly for

$$Y = NH', \quad \exp(2L^{0.35})/4 \leq H' \leq \exp(C_2 L^{0.40} \log L), \quad r \in \mathbb{N}_{>0}, \quad r \leq L^{C_0}, \quad Q = Nr, \quad (48)$$

and every $A > 0$,

$$\int_{|\beta| \leq L^{C_0}/Y} \sum_{b \bmod Q} |\widehat{a}(b/Q + \beta)|^2 d\beta \ll_{A, C, C_0, C_2} Y L^{-A}, \quad \widehat{a}(\theta) = \sum_n a(n) e(\theta n). \quad (49)$$

Consequently the energy on the union of these arcs for all $r \leq L^{C_0}$ is $O_A(Y L^{-A})$, after increasing intermediate precision.

Proof. We begin with a prime extraction that does not assume squarefreeness. For $\omega_{\mathcal{S}}(n_*) > 0$ one has the exact identity

$$-\int_0^1 \sum_{\substack{p \in \mathcal{S} \\ p \mid n_*}} \lambda(n_*/p) R_z(n_*/p) u^{\Omega_{\mathcal{S}}(n_*/p)} du = \lambda(n_*) R_z(n_*) \frac{\omega_{\mathcal{S}}(n_*)}{\Omega_{\mathcal{S}}(n_*)}. \quad (50)$$

Each integral is $1/\Omega_{\mathcal{S}}(n_*)$, and $\lambda(p) = -1$. The ratio is 1 unless the square of an $\mathcal{S}$ -prime divides n_* ; in that case the difference from the desired value has absolute value at most 1. The resulting error after all bounded factors has squared norm

$$\ll Y L^{C'} \sum_{p > \exp(L^{0.22})} p^{-2} \ll Y L^{C'} \exp(-L^{0.22}). \quad (51)$$

For counts with floors, $\sum_{p^2 \leq O(Y)} (Y/p^2 + 1) \ll Y \exp(-L^{0.22}) + \sqrt{Y}$, with the latter term absorbed. Remove this error globally by Parseval before estimating the arc union. For the grid integral at fixed Q , arcs have spacing $1/Q$ and width $O(L^{C_0}/Y)$. Since $Y/Q = H'/r$ exceeds every fixed power of L , they are disjoint for large N . The same norm bound therefore handles this integral; summing in r costs a fixed power.

After extraction the residual coefficient, for $w = n/p$, is

$$b_u(w) = \lambda(w_*) R_z(w_*) u^{\Omega_S(w_*)} \mathbf{1}_{(w,N)=1} c(wp), \quad |b_u(w)| \leq L^{C'}. \quad (52)$$

Here $p \notin \mathcal{P}$, $p \nmid N$; no condition $p \nmid w$ is imposed. Its discrepancy is exactly the square error already removed.

We need the principal residual mean. If $\log R \asymp L$, χ_0 is principal modulo $Q = Nr$, and $I \subset [R, 2R]$, then for every fixed B and $|t| \leq L^B$,

$$\sum_{w \in I} b_u(w) \chi_0(w) w^{it} \ll_A R L^{-A} \quad (53)$$

uniformly in $u \in [0, 1]$ and c . Write $w = gm$ with g $\mathcal{P}$ -smooth. The function of m is completely multiplicative: it is 0 on $\mathcal{P}$ and primes at most z , $-u$ on $\mathcal{S}$, and -1 elsewhere. Lemma 3.3 applies with $d = 0.40$. The mask $(m, N) = 1$ is supplied by χ_0 . For $g \leq \sqrt{R}$ the residual length R/g is a positive power of N , and errors sum using

$$\sum_{g \text{ } \mathcal{P}\text{-smooth}} \frac{|c(g)|}{g} \leq L^C \prod_{p \in \mathcal{P}} (1 - 1/p)^{-1} \leq L^{C'}. \quad (54)$$

For $g > \sqrt{R}$ use the trivial count and Rankin with $\delta = L^{-0.40}$. Since all group primes are at most $\exp(L^{0.40})$, the Euler product at $1 - \delta$ is a fixed power of L . The tail is $O(R L^{C'} \exp(-c_1 L^{0.60}))$. If $\mathcal{P}$ is empty the tail is zero. Differences of prefixes prove the assertion for all interval endpoints. No multiplicativity of c is needed.

Decompose p and w into dyadic intervals $p \asymp P$, $w \asymp R$, with $PR \asymp Y$. There are $O(L^{C'})$ pieces and

$$P \leq \exp(L^{0.28}), \quad R/Q \gg H'/(Pr) \longrightarrow \infty \quad (55)$$

faster than every fixed power of L . A factor 2 in a prime dyad endpoint is harmless by using a slightly wider fixed exponent range in Lemma 3.5. For a separated power twist put

$$P_\chi(t) = \sum_{p \in I_P \cap \mathcal{S}} \chi(p) p^{it}, \quad R_\chi(t) = \sum_{w \in I_R} b_u(w) \chi(w) w^{it}. \quad (56)$$

Orthogonality and the number of points in each residue class give, for every real t ,

$$\begin{aligned} \frac{1}{\varphi(Q)} \sum_{\chi \bmod Q} |R_\chi(t)|^2 &= \sum_{a \in (\mathbb{Z}/Q\mathbb{Z})^*} \left| \sum_{\substack{w \in I_R \\ w \equiv a \pmod{Q}}} b_u(w) w^{it} \right|^2 \\ &\leq (2R/Q + 1) \sum_{w \in I_R} |b_u(w)|^2 \ll R^2 L^{C'} / Q. \end{aligned} \quad (57)$$

For $|t|$ bounded by any fixed power of L , Lemma 3.5 and (57) imply

$$\frac{1}{\varphi(Q)} \sum_{\chi \neq \chi_0} |P_\chi(t) R_\chi(t)|^2 \ll \frac{Y^2}{Q} P^{-1/8} L^{C'}. \quad (58)$$

For χ_0 use $|P_{\chi_0}| \leq 2P$, (53) and $Q/\varphi(Q) \ll \log \log(3Q)$. The full character energy is consequently $O_A(Y^2 L^{-A}/Q)$, without a triangle sum over Q characters.

On a fixed product dyad, the factor $\psi(pw/Y)(pw)^{iv}e(\beta pw)$ is smooth in $\log(pw/Y)$. For $|\beta|Y \leq L^{C_0}$ its Mellin transform has integral norm bounded by a fixed power of L . Truncate at $|t| \leq L^B$ with B larger than the derivative exponents. Repeated integration by parts gives any fixed logarithmic saving in the tail. For every real t , the trivial $|P_\chi| \leq 2P$ and (57) retain the baseline $Y^2 L^{C'}/Q$, so the tail loses no factor Q . Minkowski in the normalized character Hilbert space, the u integral and the dyadic sum gives, for the extracted sequence a_0 ,

$$\frac{1}{\varphi(Q)} \sum_{\chi \bmod Q} \left| \sum_n a_0(n) \chi(n) e(\beta n) \right|^2 \ll_A Y^2 L^{-A}/Q. \quad (59)$$

The supports are units modulo Q : every prime factor of r is at most $L^{C_0} < z$, removed by roughness, and the N -unit mask removes the others. Finite additive Parseval and character orthogonality therefore give

$$\begin{aligned} \sum_{b \bmod Q} |\hat{a}_0(b/Q + \beta)|^2 &= Q \sum_{a \bmod Q} \left| \sum_{n \equiv a \pmod{Q}} a_0(n) e(\beta n) \right|^2 \\ &= \frac{Q}{\varphi(Q)} \sum_{\chi \bmod Q} \left| \sum_n a_0(n) \chi(n) e(\beta n) \right|^2 \ll_A Y^2 L^{-A}. \end{aligned} \quad (60)$$

Integrate over length $2L^{C_0}/Y$, increasing the preceding precision, and restore (51). This proves (49) for each fixed N . $\square$

Lemma 4.2 (Two-dimensional partial summation). *The smooth-factor estimate (59) also follows directly by Abel summation, with the same assumptions.*

Proof. On a prime/residual rectangle $PR \asymp Y$ define

$$A_\chi(x) = \sum_{\substack{p \in I_P \cap \mathcal{S} \\ p \leq x}} \chi(p), \quad B_\chi(y) = \sum_{\substack{w \in I_R \\ w \leq y}} b_u(w) \chi(w). \quad (61)$$

Lemma 3.5 at $t = 0$, (53) and (57) hold uniformly in these prefixes. Consequently

$$\|(A_\chi(x) B_\chi(y))_\chi\|_{\ell^2(d\chi/\varphi(Q))} \ll_A \frac{Y}{\sqrt{Q}} L^{-A}. \quad (62)$$

The nonprincipal part has the stronger factor $P^{-1/16}$ times a fixed power of L ; the principal coordinate uses any prescribed preceding precision. Set

$$F(p, w) = \psi(pw/Y)(pw)^{iv}e(\beta pw). \quad (63)$$

Its two-variable variation satisfies

$$\begin{aligned} |F(2P, 2R)| &+ \int_P^{2P} |F_p(x, 2R)| dx + \int_R^{2R} |F_w(2P, y)| dy \\ &+ \int_P^{2P} \int_R^{2R} |F_{pw}(x, y)| dy dx \ll L^{C'}. \end{aligned} \quad (64)$$

Indeed rescale $p = Ps, w = Rt$. On $[1, 2]^2$, PR/Y is in a fixed compact subset of $(0, \infty)$; derivatives through order two are bounded by a fixed polynomial in the derivative norms of ψ , $|v|$ and $|\beta|Y$. The factor $(PR)^{iv}$ has modulus 1.

With prefixes zero at the lower endpoints, two-dimensional partial summation is the exact identity

$$\begin{aligned} S_\chi &= F(2P, 2R)A_\chi(2P)B_\chi(2R) \\ &\quad - \int_P^{2P} F_p(x, 2R)A_\chi(x)B_\chi(2R) dx \\ &\quad - \int_R^{2R} F_w(2P, y)A_\chi(2P)B_\chi(y) dy \\ &\quad + \int_P^{2P} \int_R^{2R} F_{pw}(x, y)A_\chi(x)B_\chi(y) dy dx, \end{aligned} \tag{65}$$

where S_χ is the weighted double character sum. Half-open dyads can be handled by displaced real endpoints. Minkowski, (62) and (64) retain $Y/\sqrt{Q}$ while costing only a fixed power of L . Integrating u and summing the polynomially many dyads proves (59) uniformly in β . $\square$

Remark 4.3. The condition $R = Y/P \gg Q$ is essential. At $H' = 1$, the orthogonality bound contains $1 + Q/R$; with $Q \asymp N$ and $Y = N$ the nonprincipal saving becomes $P^{-1/8}(1 + Q/R) \asymp P^{7/8}$. The grid lemma is used only for comparisons with two nonempty free products, never for the raw correlation. The narrow displacement $|\beta|Y \leq L^{C_0}$ is precisely what makes (64) polynomial in L .

5 Reflected weighted cancellation

Definition 5.1 (Marked weights). For nonnegative integers $\mathbf{t} = (t_g)_g$ let $(y)_t$ denote the falling factorial and put

$$W_{\mathbf{t}}(n) = q_0^{\omega_{\mathcal{P}}(n) - \sum_g t_g} \prod_g \frac{(\omega_g(n))_{t_g}}{V_g^{t_g}}. \tag{66}$$

More generally, replace the product of falling factorials by the sum of a mark function b over the ordered distinct unshared divisor lists, retaining the same normalization and damping; write $W_{\mathbf{t}}^b(n)$. An empty admissible list set gives weight zero. If $\max t_g \leq t_{\max}$ is fixed, then

$$\sup_n |W_{\mathbf{t}}^b(n)| \leq \|b\|_{\infty} L^{C(t_{\max})}. \tag{67}$$

Indeed $\sup_{y \geq t} q_0^{y-t}(y)_t/V_g^t$ is a fixed constant for $t \leq t_{\max}$, and there are $O(\log L)$ groups. For $\ell = 1$ write

$$W_1(n) = q_0^{\omega_{\mathcal{P}}(n) - s} \prod_g \frac{\omega_g(n)}{V_g}, \quad H(n) = R_z(n) \mathbf{1}_{(n, N)=1} \mathbf{1}_{\omega_S(n) > 0} W_1(n). \tag{68}$$

Proposition 5.2 (Reflected cancellation). *Use the ranges (46), with $O(\log L)$ disjoint groups, reciprocal masses in a fixed compact subset of $(0, \infty)$, and at least $c_0 \log L$ groups in each of the small and big ranges. No group prime divides N . Assume every big group is a prime interval with the primes dividing N removed. Fix $0 < q_0 < 1$ and $\ell = 1$. For every fixed smooth ψ_0 supported in $[1/3, 2/3]$ and every fixed $A > 0$,*

$$\left| \sum_{1 \leq n < N} \psi_0(n/N) H(n) H(N-n) \lambda(n) \lambda(N-n) \right| \ll_A N L^{-A}. \tag{69}$$

The threshold is uniform over N and the indicated group construction; no average over N is taken. For odd N the rough support is empty.

Proof. Set $x = N$, take $k = 1$, and choose the residual family of Input 2.4, using Proposition 2.7. Use B_N from Proposition 2.6. We verify the lift, the individual patterns, and then their comparison terms.

At a positive physical position use, at the source and at the target, the mark-independent endpoint function

$$f(n) = \lambda(n)F_0(n)q_0^{(\omega_{\mathcal{P}}(n)-Ms)/2}, \quad F_0(n) = R_z(n_*)\mathbf{1}_{(n,N)=1}\mathbf{1}_{\omega_{\mathcal{S}}(n_*)>0}. \quad (70)$$

Both endpoints are zero outside their positive cutoffs. Every group prime exceeds z , is outside $\mathcal{S}$, and is coprime to N , so F_0 is invariant under multiplication by $\mathcal{P}$ -integers. On a physical state $\omega_{\mathcal{P}}(n) \geq Ms$, and therefore $|f(n)| \leq 1$.

For a positive dyad, summing first over ordered lists, then over multiples of their product P_{list} , gives

$$\sum_{n \asymp Y} \sum_{\text{lists at } n} \prod_g V_g^{-M} \leq CY \prod_g \left(\sum_{p \in \mathcal{P}_g} \frac{1}{pV_g} \right)^M + N^{o(1)} \ll Y, \quad (71)$$

$$\log P_{\text{list}} = O_J(L^{0.40} \log L) = o(L).$$

The rounding term is $N^{o(1)}$, since J is fixed. Thus both endpoint norms are $O(\sqrt{Y})$, with the logarithmic exponent independent of J and the probe count. Bounded smooth cutoffs and modulus-one power twists preserve this.

Insert the edge factor $\Psi(n/(ND))/n$, where

$$\Psi(t) = t\psi_0(t). \quad (72)$$

Its support makes $n' = ND - n$ positive and comparable to n . Since $D = \exp(O_J(L^{0.40} \log L))$, only a fixed power of L many size dyads are needed, and all their lengths lie between N^γ and $N^{1/\gamma}$ for a fixed $\gamma > 0$.

Choose a smooth dyadic partition of unity and smooth target cutoffs equal to 1 on the corresponding comparable positive target ranges. If the source scale is Y , the source and target cutoff derivatives of order j are $O_j(Y^{-j}L^{C(j+1)})$. They may be obtained by rescaling fixed smooth functions; the logarithmic derivative exponents are fixed before the graph parameters. Overlapping size intervals may each be split into finitely many support dyads for transference. The smooth partition and target cutoffs sum back to the original expression on the support of Ψ before any individual pattern is evaluated. All subsequent cutoffs used for Lemmas 4.1 and 4.2 have these same derivative bounds.

For $\Phi(u) = \Psi(\exp u)$ and $\widehat{\Phi}(t) = \int_{\mathbb{R}} \Phi(u) \exp(-itu) du$, logarithmic Fourier inversion is

$$\Psi(n/(ND)) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\Phi}(t)(n/N)^{it} D^{-it} dt. \quad (73)$$

On a source dyad place Y/n and the conjugated source twist in the first endpoint, and divide the pairing by Y . The allowed ideal frequency is $\zeta = -t$. The transform has bounded or fixed-log-power integral norm. If its logarithmic derivative exponent is C_ψ , fix $F > C_\psi + 2$ in advance and take $C_4 = F$ in Input 2.4, so that every frequency $\zeta = -t$ with $|t| \leq L^F$ is admissible there. Integration by parts gives

$$\int_{|t| > L^F} |\widehat{\Phi}(t)| dt \ll_j L^{C_\psi(j+1)-F(j-1)}. \quad (74)$$

Arbitrarily strong saving follows by increasing j without changing F or C_4 . By Propositions 2.5 and 2.6, the absolute row and column sums of B_N are at most $L^{C_{\text{abs}}}$ for every ζ , so each pairing is $O(YL^{C_{\text{abs}}})$ by the endpoint norms; together with (74) this disposes of the tails. Choose the residual precision, then E_{id} , then the probes and kernel precisions, then sufficiently large fixed J , and finally N . Propositions 2.5–2.7 and Input 2.4 make the entire normalized signed residual pairing $O(L^{-B})$ for any prescribed B .

We now compute its individual patterns. This uses the normalization of [S, Section 5.2, pp. 36–38] and the shared-sign identity of [E, (4.18)–(4.19), p. 16], with reflection included explicitly. In group g put $c_g = |C_g|$, $s_{R,g} = J - c_g$, $t_g = 1 + c_g$, so $M = s_{R,g} + t_g$. Write $D = D_R D_C$ for shared and auxiliary free products. Shared labels divide both endpoints; hence

$$n = D_R w, \quad n' = ND - n = D_R w', \quad w + w' = ND_C. \quad (75)$$

The cutoff and harmonic normalization become exactly

$$\frac{\Psi(n/(ND))}{n} = \frac{\psi_0(w/(ND_C))}{ND_R D_C}. \quad (76)$$

The state and target-list normalizations satisfy

$$V_g^{-M-t_g} = V_g^{-s_{R,g}-2t_g} = V_g^{-s_{R,g}} V_g^{-t_g} V_g^{-t_g}. \quad (77)$$

The factor $1/D_R$ converts every shared-label sum with normalization V_g^{-1} into a probability draw $1/(pV_g) = \mu_g(p)$, leaving $V_g^{-t_g}$ at each endpoint.

Away from shared overlaps,

$$\omega_g(D_R w) - M = \omega_g(w) - t_g, \quad \omega_g(D_R w') - M = \omega_g(w') - t_g. \quad (78)$$

At each endpoint the graph half-damping in (8) and the endpoint half-damping in (70) give $q_0^{\omega_{\mathcal{P}}(n)-Ms}$. By (78) these are exactly the full powers $q_0^{\omega_{\mathcal{P}}(w)-\sum_g t_g}$ and $q_0^{\omega_{\mathcal{P}}(w')-\sum_g t_g}$ in the two marked weights. The sign identity, valid even on overlaps, is

$$\lambda(D_R w) \lambda(D_R w') = \lambda(D_R)^2 \lambda(w) \lambda(w') = \lambda(w) \lambda(w'). \quad (79)$$

Also $F_0(D_R w) = F_0(w)$ and $F_0(D_R w') = F_0(w')$. Comparison coefficients use free and designated unshared big labels, never shared labels, so they are unchanged by summing the shared draws. The endpoint vectors are mark-independent, so symmetrization is an average that does not alter this calculation.

We justify replacement by unrestricted shared draws in absolute value. Given w, w', D_C and preceding shared draws, at most $O_J(L)$ primes are forbidden: the prime divisors of the two endpoints, free labels and preceding shared labels. Every draw has atom $O(\exp(-L^{0.12}))$; there are $O_J(\log L)$ draws. For distinct shared values, the exact overlap identity is

$$\omega_{\mathcal{P}}(D_R w) - Ms = \omega_{\mathcal{P}}(w) - \sum_g t_g - \#\{p : p \mid D_R, p \mid w\}. \quad (80)$$

Ignoring the overlap damping loses at most $q_0^{-2Ms} = L^{O_J(1)}$ across the two endpoints. Unshared divisor lists still divide w, w' . Their bounded marked weights in (67), $|F_0| \leq 1$, and harmonic normalization bound the remaining absolute sum by a fixed power of L . The total shared-exclusion error is

$$L^{O_J(1)} \exp(-L^{0.12}). \quad (81)$$

If a banned free prime p divides an endpoint, then $p \mid D_C$ and (75) imply it divides both w, w' . At scale $Y \asymp ND_C$, summing these multiples costs $O(YL^C/p)$, or $O(L^C/p)$ after normalization. Indeed $w = pv$, $w' = p(ND_C/p - v)$; both ranges are positive and have length $O(Y/p)$, with $Y/p = N^{1+o(1)}$. There are only $O(\log L)$ free slots with fixed probe count and $p \geq \exp(L^{0.35})$, so this error is also smaller than every fixed negative power of L . No cancellation is used for deleted terms.

After these replacements the pattern is

$$\frac{1}{N} \mathbb{E}_{\text{free}} \frac{1}{D_C} \sum_{\substack{w > 0 \\ w' = ND_C - w > 0}} \psi_0(w/(ND_C)) F_0(w) F_0(w') \lambda(w) \lambda(w') W_{\mathbf{t}}(w) W_{\mathbf{t}}(w') \mathbb{E}_{\text{marks}} K_{\nu}. \quad (82)$$

The inner expectation averages independent uniform ordered unshared divisor lists, giving zero if either list set is empty. The raw pattern has $D_C = 1$, $t_g = 1$, $K_0 = 1$, and gives exactly

$$\frac{1}{N} \sum_{1 \leq w < N} \psi_0(w/N) H(w) H(N-w) \lambda(w) \lambda(N-w). \quad (83)$$

For each comparison expand the two cell/character kernels in (14) and separate the smooth factors logarithmically. Up to a total fixed power of L in coefficients and integral norms, a term indexed by a pair of cells (j, j') is

$$\frac{1}{Y} \mathbb{E}_{D_A, D_B} b_A(D_A) b_B(D_B) \sum_w U(w) V(ND_A D_B - w), \quad |b_A|, |b_B| \leq 1. \quad (84)$$

For this term let U_A, U_B be the lower multiplicative size endpoints of the two cells, and put $H' = U_A U_B$, $Y = NH'$. Explicitly, the term belongs to primitive characters χ, χ' of conductor at most Q_0 and to a real frequency τ , $|\tau| \leq L^F$, of the expansion (73) applied to ψ_0 in place of Ψ , and

$$b_A(D) = \mathbf{1}_{\log D \in I_j} \overline{\chi(D)} \frac{U_A}{D} D^{-i\tau}, \quad b_B(D) = \mathbf{1}_{\log D \in I_{j'}} \overline{\chi'(D)} \frac{U_B}{D} D^{-i\tau}, \quad (85)$$

$$U(w) = \Xi(w/Y) (w/N)^{i\tau} \lambda(w) F_0(w) W_{\mathbf{t}}^{b_s}(w), \quad V(w') = \Xi(w'/Y) \lambda(w') F_0(w') W_{\mathbf{t}}^{b_t}(w').$$

Here Ξ is a fixed smooth function with compact support in $(0, \infty)$ and $\Xi = 1$ on $[1/3, 2e^2/3]$, the source mark function is $b_s = \mathbf{1}_{\log X_B \in I_{j'}} \chi'(X_B)$, and the target mark function is $b_t = \mathbf{1}_{\log X_A \in I_j} \chi(X_A)$. On the two cells $D_A D_B \in [H', e^2 H']$, since $\mathfrak{h} \leq 1$. As ψ_0 is supported in $[1/3, 2/3]$, on the support of $\psi_0(w/(ND_A D_B))$ both w and $w' = ND_A D_B - w$ lie in $[Y/3, 2e^2 Y/3]$, so both cutoffs equal 1 there. Moreover $1/(ND_A D_B) = Y^{-1} (U_A/D_A) (U_B/D_B)$. The term carries the coefficient $-(-1)^{|A|+|B|} \widehat{\Phi}_0(\tau) / (2\pi \nu_j \nu_{j'})$, where $\Phi_0(u) = \psi_0(\exp u)$ and $\nu_j, \nu_{j'} \geq \xi$ are the two cell probabilities; the frequencies $|\tau| > L^F$ are discarded as in (74), using $\|U\|_2 \|V\|_2 \ll YL^C$. The free products are independent, nonempty, and each contains a big prime. Thus

$$\exp(2L^{0.35})/4 \leq H' \leq \exp(C_2 L^{0.40} \log L), \quad \rho_A(d), \rho_B(d) \ll L^C/d. \quad (86)$$

The harmless factor $1/4$ covers endpoint conventions. The mass bounds are (23), with at most the fixed probe count per group. Both endpoint sequences have squared norm $O(YL^C)$. They have exactly the form (47): the group coefficient is $\lambda(g)$ times its bounded marked weight and big-mark kernel factors. These depend only on $g = n\mathcal{P}$. The first endpoint retains a reserved $\mathcal{S}$ -prime.

Handle the major arcs separately for this particular (j, j') and these U, V, Y . Fourier inversion supplies

$$\mathcal{M}_{j,j'}(\theta) = \mathbb{E} b_A(D_A) b_B(D_B) e(-N\theta D_A D_B). \quad (87)$$

Apply Dirichlet approximation to $N\theta$ with maximum denominator $\lfloor H'/L^B \rfloor$. For its reduced fraction u/r the error is $O(L^B/(rH'))$ and at most r^{-2} . If $r > L^B$, the bilinear estimate [S, Lemma 4.2, (4.5)–(4.6), pp. 28–29] and (86) give arbitrary fixed logarithmic saving. More explicitly, its squared normalized bound is a fixed power of L times

$$\frac{1}{U_B} + \frac{1}{r} + \frac{\log(2r)}{U_A} + \frac{r \log(2r)}{H'}.$$

Both product sizes exceed $\exp(L^{0.35})/2$; the log factor is a fixed power of L , and $r \leq H'/L^B$. Increasing B therefore gives $|\mathcal{M}_{j,j'}| \ll L^{-D}$ for any prescribed D . For $r \leq L^B$ one has

$$\theta = \frac{b}{Nr} + O\left(\frac{L^B}{NrH'}\right), \quad (88)$$

so the major set $\mathfrak{M}_{j,j'}$ is contained in the grids of Lemma 4.1 with width at most L^B/Y . Its many rational centers are retained in that lemma.

On the complement of $\mathfrak{M}_{j,j'}$, Parseval bounds (84) by L^{-D+C} . On $\mathfrak{M}_{j,j'}$ use $|\mathcal{M}_{j,j'}| \leq 1$, Lemma 4.1 for U , and global Parseval for V :

$$\frac{1}{Y} \int_{\mathfrak{M}_{j,j'}} |\widehat{U}(\theta)\widehat{V}(\theta)| d\theta \ll_A L^{-A/2+C}. \quad (89)$$

Reflection only changes the Fourier conjugation convention, not this absolute bound. Choose the grid precision after all fixed comparison costs, and sum these estimates term by term over the cell pairs, characters and separated smooth integrals. Subtract the comparisons from the small residual pairing and restore the two exclusion errors and Fourier tails. The raw expression (83) proves (69).

Finally, H depends only on the fixed groups, q_0 and $\ell = 1$, not on J . Increasing accuracy changes the auxiliary graph and threshold but leaves this weight fixed. The comparison coefficient exponents, set by the probes, are paid afterward by the arbitrary-precision grid lemma. No choice of J feeds back into the ideal norm target. $\square$

6 A bounded weight with positive rough-pair mass

Proposition 6.1 (Weight construction and moments). *In each range $[\exp(L^{0.12}), \exp(L^{0.18})]$ and $[\exp(L^{0.35}), \exp(L^{0.40})]$, list the primes not dividing N in increasing order and close a group when its reciprocal mass reaches 10. Discard the last incomplete group in each range. Then*

$$10 \leq V_g \leq 10 + \exp(-L^{0.12}), \quad (90)$$

$$|\mathcal{G}_{\text{small}}| = 0.006 \log L + O(1), \quad |\mathcal{G}_{\text{big}}| = 0.005 \log L + O(1).$$

Use $q_0 = 9/10$, $\ell = 1$ and (68). Fix a smooth $0 \leq \psi_0 \leq 1$ supported inside $[1/3, 2/3]$ with $I_\psi = \int \psi_0(t) dt > 0$. For sufficiently large even N ,

$$\sum_{1 \leq n < N} \psi_0(n/N) H(n) H(N-n) \geq cNL^{-59/250}, \quad 59/250 = 0.236. \quad (91)$$

For every fixed $A > 0$ the two one-sign moments satisfy

$$\sum_{1 \leq n < N} \psi_0(n/N) H(n) H(N-n) \lambda(n) = O_A(NL^{-A}),$$

$$\sum_{1 \leq n < N} \psi_0(n/N) H(n) H(N-n) \lambda(N-n) = O_A(NL^{-A}). \quad (92)$$

Moreover $0 \leq H(n) \leq 1$ for every n .

Proof. Mertens' harmonic prime estimate and (22) give (90) uniformly in N . The overshoot at closing a group is at most the largest reciprocal prime in its range, and the total overshoot is $o(1)$. For large N , $s \leq 0.012 \log L$ and each family contains at least $0.004 \log L$ groups. Thus all graph inputs hold with the fixed choices

$$\begin{aligned} a &= 0.12, & b &= 0.18, & c &= 0.35, & d &= 0.40, \\ c_0 &= 0.004, & C_0 &= 0.02, & v_- &= 10, & v_+ &= 11, & q_0 &= 0.9, & \ell &= 1. \end{aligned} \quad (93)$$

Every big group is an interval with the primes dividing N removed. All these data are fixed before N ; only allowed prime sets, group counts and the source moment length grow.

For $k \geq 1$,

$$kq_0^{k-1} \leq 1 + q_0 + \cdots + q_0^{k-1} \leq (1 - q_0)^{-1} = 10 \leq V_g. \quad (94)$$

Each local factor in W_1 is at most 1, and a missing group gives 0. The remaining factors of H are indicators.

Temporarily omit the N -unit masks and use independent uniform residues at each prime. The rough factor is

$$V_N(z) = \prod_{p \leq z} (1 - \nu_N(p)/p), \quad \nu_N(p) = \begin{cases} 1, & p \mid N, \\ 2, & p \nmid N. \end{cases} \quad (95)$$

For even N the factor at 2 is $1/2$. For $p \geq 3$, use $1 - 2/p$ as a lower bound. Its ratio to $(1 - 1/p)^2$ has a positive convergent product, so $V_N(z) \geq c/(\log z)^2 = cL^{-1/5}$ uniformly in N .

At a group prime $p \nmid N$, the two endpoint divisibility indicators cannot both be 1. The degree generating function is

$$J_g(u, v) = \prod_{p \in \mathcal{P}_g} \left(1 + \frac{u-1}{p} + \frac{v-1}{p} \right). \quad (96)$$

The local expectation of the two W_1 factors is exactly

$$\begin{aligned} h_g &= V_g^{-2} \partial_u \partial_v J_g(q_0, q_0) \\ &= \frac{J_g(q_0, q_0)}{V_g^2} \left\{ \left(\sum_{p \in \mathcal{P}_g} \frac{1}{p + 2q_0 - 2} \right)^2 - \sum_{p \in \mathcal{P}_g} \frac{1}{(p + 2q_0 - 2)^2} \right\}. \end{aligned} \quad (97)$$

Since $V_g = 10 + o(1)$ uniformly and the smallest group prime tends to infinity,

$$h_g = \exp(2(q_0 - 1)V_g)(1 + o(1)) = e^{-2}(1 + o(1)) \geq e^{-3} \quad (98)$$

for all groups at sufficiently large N . The error is $O(\sum_{p \in \mathcal{P}_g} p^{-2})$, with constants uniform for $V_g \in [10, 11]$.

The reserved-prime condition at both endpoints has factor

$$B_S = 1 - 2 \prod_{p \in S} (1 - 1/p) + \prod_{p \in S} (1 - 2/p) \geq 1 - 2 \prod_{p \in S} (1 - 1/p) \geq \frac{1}{2}. \quad (99)$$

Here $\sum_{p \in S} 1/p = 0.06 \log L + O(1)$. The rough primes, graph groups and reserved primes are disjoint. Their independent mean is consequently

$$\mathfrak{M}_N = V_N(z) B_S \prod_g h_g \geq cL^{-1/5} e^{-3s} \geq cL^{-1/5-0.036} = cL^{-59/250}. \quad (100)$$

We now justify a finite residue comparison strong enough to multiply by signs. Each endpoint group factor expands exactly as

$$\frac{1}{V_g} \sum_{p \in \mathcal{P}_g} \mathbf{1}_{p|n} \prod_{\substack{p' \in \mathcal{P}_g \\ p' \neq p}} (1 - (1 - q_0) \mathbf{1}_{p'|n}), \quad (101)$$

and likewise for $N - n$. The selected mark primes at the two endpoints must be distinct because $p \nmid N$. There are $2s$ selected primes in every nonzero term, with product at most $\exp(O(L^{0.40} \log L))$. The sum of their normalized independent congruence densities is at most $\prod_g (V_g/V_g)^2 = 1$.

Write each reserved-prime indicator as 1 minus its absence product and expand into four terms. Conditional on any selected mark tuple, each resulting unsigned avoidance product has factors $1 - X_p$, where $0 \leq X_p \leq 1$, and independent expected sum $\sum_p \mathbb{E} X_p = O(\log L)$ uniformly in that tuple. At group primes the two endpoint predicates are disjoint. At a rough prime dividing N the coincident predicates are counted only once. Evaluate selected-prime factors before expanding. Bonferroni truncations at consecutive orders $h, h+1$, with h an even integer exceeding $C_A \log L$, sandwich these products. The independent expected gap is at most

$$\frac{(C \log L)^{h+1}}{(h+1)!} \leq L^{-A} \quad (102)$$

after C_A is fixed sufficiently large.

Every retained term is a residue class, or a union expanded into at most 2^{h+1} residue classes, modulo a squarefree product of at most $2s + h + 1$ primes at most $\exp(L^{0.40})$. The total absolute coefficient mass, including selected marks and residue choices, is

$$\exp(O_A(L^{0.40} \log L)) = N^{o(1)}. \quad (103)$$

For each residue class modulo d , smooth counting gives $NI_\psi/d + O(1)$, uniformly in the residue and d . One way to see the $O(1)$ is to integrate the bounded discrepancy of a progression against the fixed function ψ_0 . The finite polynomial average therefore equals its independent average with error $N^{-1+o(1)}$. Combining the sandwich bounds and (102) gives average error $O_A(L^{-A}) + N^{-1+o(1)}$. More strongly, choose one sandwich polynomial for each of the four unsigned pieces. The nonnegative gap controls its absolute error. Hence the exact nonnegative weighted product, without the N -unit masks, has a finite residue polynomial approximation whose L^1 error against $\psi_0(n/N)$ is $O_A(NL^{-A})$. The four pieces cost only a factor four.

Restoring the N -unit masks changes the bounded weighted sum only if a prime $p \mid N$, $p > z$, divides an endpoint. Its total contribution is at most

$$2 \sum_{\substack{p|N \\ p > z}} (N/p + 1) \ll \frac{NL}{z \log z} + \frac{L}{\log z}, \quad (104)$$

smaller than every NL^{-A} . Thus the unsigned moment is $NI_\psi \mathfrak{M}_N + O_A(NL^{-A})$, which proves (91).

For a one-sign moment, remove the unit masks from the exact bounded weight using (104), before passing to the residue polynomial. Multiply its L^1 approximation by $\lambda(n)$. All retained moduli are $N^{o(1)}$. For a class $n \equiv a \pmod{d}$, put $g = (a, d)$. Writing $n = gu$ leaves a reduced class modulo d/g and $\lambda(gu) = \lambda(g)\lambda(u)$. Character orthogonality in the reduced class and Lemma 3.2, with a fixed margin and partial summation for ψ_0 , give $O(N^{15/16+o(1)})$ per retained class. The total coefficient mass (103) makes the aggregate $O(N^{0.99})$ for sufficiently large N . Add the approximation error at arbitrary fixed precision. The second one-sign moment follows by reflection and the cutoff $\psi_0(1-t)$. This proves (92). $\square$

7 Proof of the two theorems

Proof of Theorem 1.1. Take H, ψ_0 from Proposition 6.1 and set

$$T_{\sigma, \tau} = \sum_{1 \leq n < N} \psi_0(n/N) H(n) H(N-n) \frac{1 + \sigma \lambda(n)}{2} \frac{1 + \tau \lambda(N-n)}{2}. \quad (105)$$

Each summand is nonnegative, at most 1, and supported on the set in (1). Expanding the sign indicators gives one quarter of the unsigned moment and the two one-sign moments in Proposition 6.1, and one quarter of the double-sign moment in Proposition 5.2. For example, use precision $A = 2$. Uniformly in all four signs,

$$T_{\sigma, \tau} \geq c_1 N L^{-59/250} - C_1 N L^{-2} \geq c_2 N L^{-59/250}. \quad (106)$$

Since $59/250 < 1/4$, the unweighted count is at least the bound in (1). All constants and the threshold are uniform in even N and the sign pair. $\square$

Remark 7.1. The parity information uses the graph residual estimate, complete multiplicativity in (79), and the prime and all-character estimates behind the grid lemma. Unsigned residue data alone do not bound the double-sign moment. Replacing every sign by $+1$ would fail (53). For a completely multiplicative sign with $f(p) = (p/q)$ for $p \neq q$, $f(q) = -1$, and $q \equiv 3 \pmod{4}$ prime, where $(\cdot/q)$ is the Legendre symbol, a unit pair at an even total divisible by q satisfies

$$f(n)f(N-n) = (n/q)(-n/q) = -1. \quad (107)$$

The shared-factor identity still holds, but the reserved prime values are not constantly -1 and twisting by the character modulo q violates (35). Thus the general sign function fails at the prime extraction and principal residual step. The all-modulus zero-free input also excludes any real zero above $7/8$ that would obstruct the character contours. No character at the grid modulus is suppressed.

Roughness alone gives, for either represented $a \in \{n, N-n\}$,

$$\Omega(a) \log z < \log a < \log N, \quad \Omega(a) < (\log N)^{9/10}. \quad (108)$$

Thus $\lceil (\log N)^{9/10} \rceil$ already bounds both factor counts with each prescribed parity. The next argument gives the smaller growing bound (2).

Proof of Theorem 1.2. The number of z -rough nonsquarefree integers $a \leq N$ is at most

$$\sum_{p > z} \left\lfloor \frac{N}{p^2} \right\rfloor \leq N \sum_{m > z} m^{-2} \leq \frac{N}{z-1}. \quad (109)$$

For squarefree a , the divisor function satisfies $d(a) = 2^{\Omega(a)}$. The elementary divisor identity gives

$$\sum_{a \leq N} d(a) = \sum_{m \leq N} \lfloor N/m \rfloor \leq N(1 + \log N). \quad (110)$$

The number of squarefree $a \leq N$ with $\Omega(a) > K(N)$ is at most

$$2^{-K(N)} N(1 + \log N) \ll N L^{1-2 \log 2}. \quad (111)$$

Apply both bounds to each endpoint via $a \mapsto N - a$. The total number of discarded representations, irrespective of sign, is

$$\ll N/(z-1) + NL^{1-2\log 2} = o(NL^{-1/4}), \quad (112)$$

since $\log 2 > 2/3$ and $1 - 2\log 2 < -1/3$. Subtract this unsigned error separately from each of the four lower bounds in Theorem 1.1. At least half its stated mass remains for large N , giving the asserted possibly smaller constant. $\square$

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Use of AI

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